Shreya Gunjkar
1032240176

MPJ ASSIGNMENT 5

Aim - Develop a web application for Employee Registration using JSP, Servlet and JDBC

Objectives :- To understand server side scripting
→ To understand concept of Servlet and JSP
→ To learn working of Servlet and JSP
→ To learn database connectivity using MySQL
→ To develop web application using Servlet, JSP and database connectivity

Theory -

1. Servlet
→ It is a Java program that runs on a Java enabled web server or application server.
→ It handles client requests, processes them and generates responses dynamically.
→ It is the backbone of many server side java applications due to their efficiency and scalability.

2. JSP (Javaserver Pages)
→ It creates dynamic web applications
→ It allows developers to embed Java code directly into HTML pages and it makes web development more efficient.
→ It is platform independent, write once run anywhere

3. Database Connectivity
→ It is achieved in Java using Java Database Connectivity (JDBC) API.
→ It allows java programs to connect to a database, run queries, retrieve and manipulate data.

Platform

- JSP - 2.2+
- IDE - Eclipse Neon.3
- JDK - 1.8 or later
- Apache Tomcat - 8.5
- JSTL - 1.2.1
- Servlet API - 2.5
- MySQL - mysql-connector-java-8.0.13.jar

Conclusion - Thus, studied and implemented servlets, JSP and database connectivity.

FAQs:

Q1.) What is the difference between servlet and JSP?

Ans

Servlet	JSP
→ We write HTML in servlet code	→ We write JSP in HTML.
→ Faster than JSP	→ Slower than servlet
→ Writing code harder	→ Writing code is easier.
→ We cannot run java script at client side	→ We can run java script at client side

Q2.) List out advantages of using JSP.

Ans
→ It is platform independent
→ It simplifies web development
→ User need not write HTML and Java code separately.
→ It is efficient
→ It is scalable and can easily integrate with other backend services.

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Q3.) What is the difference between Get and Post methods?

Ans

Get	Post
→ Only limited amount of data can be sent because data is sent in header	→ Large amount of data can be sent because data is sent in body
→ Not secure	→ Secured
→ Can be bookmarked	→ Cannot be bookmarked
→ Often cacheable	→ Hardly cacheable

Q4.) Compare GenericServlet and HttpServlet.

Ans

GenericServlet	HttpServlet
→ Protocol independent	→ Protocol dependent
→ We cannot define different logic for get() & post() method	→ We can define different logic for get() & post() method
→ Redirection not possible	→ Redirection possible
→ Cannot do session management with cookies or http session	→ Can do session management with cookies and http session.

Q5.) What are the major components of JDBC?

Ans 1) Application - It can be a Java application or servlet with a data source

2) JDBC API - Allows programs to execute SQL queries and get results from database.

3) DriverManager - Uses some database specific drivers to effectively connect enterprise applications to databases

4) JDBC drivers - These drivers handle interaction between application & database.